Hotel Management System

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Project Description:

Hotel management involves ensuring the smooth flow of an establishment by keeping records of sales, staff, and reservations. A management system that can keep track of those items can facilitate the flow of the hotel. With the increasing popularity of concerts, people travelling more since Covid-19, and the internet making it easier to book rooms and gain points for better deals and tiered rewards, hotels have become increasingly popular.

This Hotel Management System will help the hospitality industry to manage their sales and compare their profit to loss. It will help keep track of cost and inventory, what staff was working during those sales or reservations, what rooms were booked, which rooms were dirty, the occupancy of the restaurant, what sold in the small hotel market to track expenses. For instance, if employees A and B worked the bar and kitchen respectively, they would be on the system, and the food and drinks would be during their schedule, and that way they could calculate tips and sales. Liquor weight could also be accounted for, some hotels weigh their liquor to keep track of sales and inventory after shifts and to check for disparities.

This project is intended to create a smoother flow of operations during hotel shifts, and make it easier for hotel owners to keep track of their sales, hours and inventory in order to maximize their profit. It could help keep track of their peak seasons, and adjust appropriately for future decisions.

Objectives:

- Collect records of hotel reservations, sales and staff that include date of reservations, room cost, food sales, drink sales, market kiosk sales, occupancy, and customer names.
- Identify what dates and items sell the most.
- Identify what liquors is the most sold to encourage new cocktail recipes with those brands.
- Identify what food items sell the most for inventory and encourage different variations of items in the restaurant.
- Identify what items do not sell to replace them off the menu or market sales.
- Keep track of customer records to encourage regular customer-employee relationships.
- With those regular customers, encourage them to spend more time at the hotel to gain points in the system so they can upgrade their tiers, for example from silver to gold member.
- Analyze trends in the seasons with real life occasions, for example, college move-in dates, or concert or wedding seasons to see if there are more people spending time at the hotel and making use of amenities or not making use of them at all.

- Identify food and beverage employees' average sales to keep track of "best" employees.
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Scope:

1. The primary role of the Hotel Management system is to encourage better decision making and identify trends in sales and reservations.
2. The centralization of data includes name, confirmation/itinerary number, revenue from that day, staffs working, room charge, dates, tips, room number, and item sold.
3. Combining different modules addressing reservation management, revenue, inventory and items sold, and staff management to ensure a smooth hotel shift and analyze trends that can help generate profit.
4. Generate automated reports on inventory will automatically be updated whenever an item is sold due to self-kiosk and pos system.
5. Computerizes the inventory of products from the market, food and beverage using POS to create inventory reports.
6. Backs up the system in case of loss of data.
7. Generate reports on frequency of items bought, days that have high occupancies, staff working when there is a high average revenue, the changes in room rates, frequency of group hotel reservations.

Project Requirements:

Operating System: Windows
Database: MySQL
Applications: MYSQL Workbench, Lucidchart, Microsoft Word

User Requirements:

- The system generates a unique id per guest and stores guest information, for example, reservation on the customer column.
- The system generates a confirmation number for each guest reservation from the hotel website or an itinerary number from third party websites.
- The system records whether the customer has hotel membership, and if they do, what tier they are.
- Upon checking out after their stay, points will be added to their account if they have a member account.
- Silver members get free water, Gold member customers get a 10 usd spending per guest, Platinum members get a 15 usd spending per guest, and diamond members get a 20 usd spending per guest that will need to be charged to their room if they want to spend it.

- When there are charges to the room, the system will record the money spent at the end of their stays.
- If the customer uses card or cash, it will go to miscellaneous revenue tab instead of being assigned a name.
- The weekly reports record the date, confirmation number/itinerary number, money spent by the confirmation number during their stay (room charge), name of customer, staff working during those hours, miscellaneous revenue for each day and tips.
- Beverage staff at the end of their shift records their tip amount in daily tip journal and breakfast records their tips made as well in order for management to calculate shared tips.
- Management staff is responsible for the recording of any changes that might happen in the data.

Database Requirements:

The following is a list that provides the different data tables that will be used.

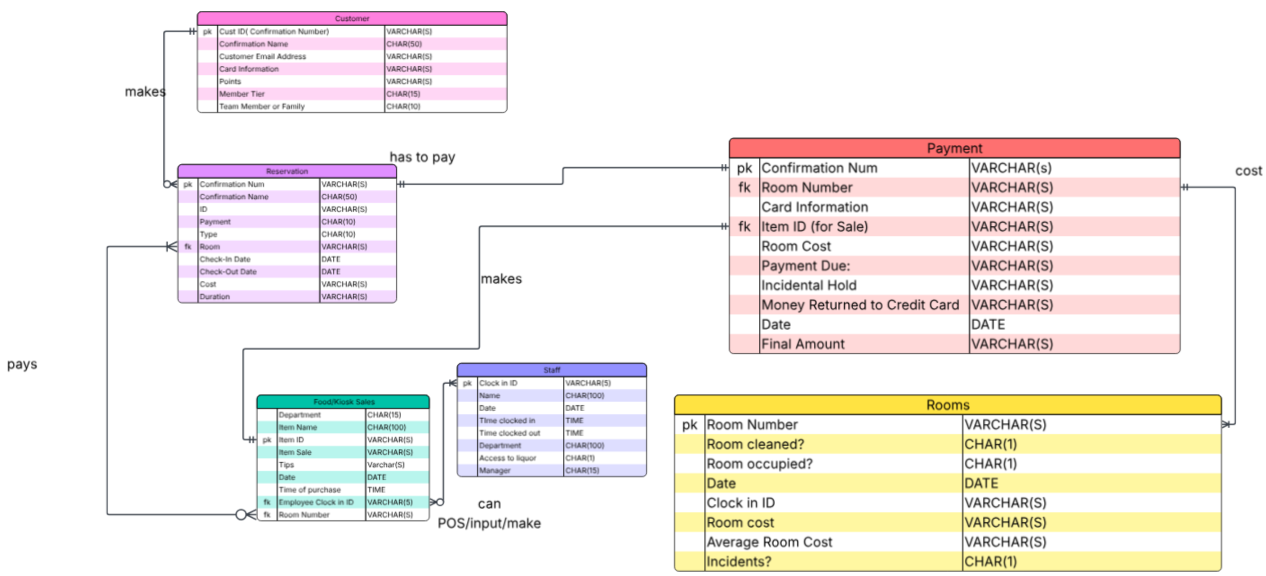
1. Reservation Table
2. Items Sales Table
3. Staff Table
4. Payment Table
5. Rooms Table

Business Rules:

1. Every customer has a unique ID
2. Every customer must have check in date and check out date.
3. Customers do not get free breakfast unless they are Gold tier and above members.
4. Diamond members are offered free parking closer to the entrance.
5. Groups are allowed to get a 10% discount rate for their stay, as well as free breakfast.
6. Rooms reservations can be placed through the website, 3rd party websites and by calling.
7. The hotel has no control over 3rd party promises and will not be held liable for any benefits made by the 3rd party website that was not stated on the reservation.
8. $50 and $100 bills should not be accepted for small totals, direct customers to the nearest ATM.
9. Accounts can only be associated with one customer.
10. Every item should have a unique ID number.
11. Team members get a discount rate depending on occupancy.

12. Bar and breakfast staffs handling cash should put the money made from cash sales in an envelope and record it at the end of their shifts.
13. Sales made from food and beverage should be recorded in the cash book at the end of each shifts.
14. Only the manager on site during the restaurant shift should have the keys to the liquor room.
15. Customer email address should be unique.
16. Customer information must be recorded before making a reservation.
17. Rooms cleaned should be recorded before assigning a customer a room.
18. The room rate increases with increased occupancy.
19. An incidental hold must be placed on the credit card of the customer.

Entity Relationship Diagram (ERD)



Data Dictionary:

Table Name	Attribute Name	Description	Data Type	Data Format	Required	PK or FK	Example
	Confirmation_Num	The unique number associated with the stay	VARCHAR(S)	9999999999	Y	PK	3292943337
CUSTOMER	Customer Name	The name of the customer	CHAR(50)	Xxxxxxxx	Y		Marylyn Wolf
	Customer Email Address	The email of the customer	VARCHAR(S)	XXXXXX@xxxxx.xxx	Y		Marylynwolf@yahoo.com
	Card Information	Details on the card	VARCHAR(S)	9999999999	Y		7373 82737 83838
	Points	The amount of pointd accumulated	VARCHAR(S)	99999			12030
	Member tier	Reward Tier of the Customer	CHAR(15)	Xxxxxxxx			Silver
	Team Member or Family						
Reservation	Confirmation_Num	The unique number associated with the stay	VARCHAR(S)	9999999999	Y	PK	3292943337
	Customer Name	The name of the customer	CHAR(50)	Xxxxxxxx	Y		Marylyn Wolf
	ID	Government Identification of Customer	VARCHAR(S)	99999999	Y		98371273
	Payment	Type of Payment	CHAR(10)	Xxxxxxxx	Y		Card
	Type	Group or Individual	CHAR(10)	Xxxxxxxx			Individual
	Room	The room number	VARCHAR(S)	999	Y	fk	324
	Check-In Date	Date when the customer gets here	DATE	mm-dd-yyyy	Y		10/9/2024
	Check-out Date	Date when the customer leaves	DATE	mm-dd-yyyy	Y		10/15/2024
	Duration	How many days the customer is staying	VARCHAR(S)	99	Y		4
	Cost	The amount paid for the stay	VARCHAR(S)	99999	Y		800
Staff	Clock in ID	Employee clock in ID	VARCHAR(S)	99999	Y	pk	
	Name	Name of Employee	CHAR(100)	Xxxxxxxx	Y		Polly Pocket
	Date	Date of work schedule	DATE	mm-dd-yyyy	Y		10/8/2025
	Time clocked in	Time employee clocked in	TIME	hh:mm	Y		07:00
	Time clocked out	Time employee clocked out	TIME	hh:mm	Y		03:00
	Type	The type of employee	CHAR(100)	Xxxxxxxx	Y		Front Desk
	Access to liquor?	Does the employee have access to liquor	CHAR(1)	X			No
	Manager	Manager working at the time	CHAR(15)	Xxxxxxxx			Chris Rules
Food/Kiosk Sales	Department	Kiosk or Restaurant?	CHAR(15)	Xxxxxxxx	Y		Restaurant
	Item Name	The Name of the Item puchased by customer	CHAR(100)	Xxxxxxxx	Y		Martini
	Item ID	The ID of the Item puchased by customer	VARCHAR(S)	9999999	Y	PK	929299
	Item sale	The sale by the item	VARCHAR(S)	9999	Y		12
	Tips	Tips made by the sale	VARCHAR(S)	30.02			
	Date of Purchase	The Date the purchases was made	DATE	mm-dd-yyyy	Y		9/30/2025
	Time of Purchase	Time purchases was made	TIME	hh:mm	Y		09:09
	Employee Clock in ID	The clock in ID of the employee	VARCHAR(S)	999999	Y	FK	23459
	Room Number	The number of the customer's room	VARCHAR(S)	999	Y	FK	98
Payment	Confirmation Num	The unique id for the puirchase	VARCHAR(S)	99999999	y	PK	849392
	Room Number	The number of the customer's room	VARCHAR(S)	9999	y	FK	234
	Card information	The information on the customer's card	VARCHAR(S)	9999999	y		9239 928992
	Item ID(For Sale)	The unique id for the puirchase	VARCHAR(S)	9999999			4532
	Room Cost	The cost of the customer's room	VARCHAR(S)	999		FK	129.38
	Payment Due:	The payment due for the customer	VARCHAR(S)	99999	y		938.92
	Incidental Hold	A hold on the card in case something breaks	VARCHAR(S)	999			50.93
	Money Returned to Credit C	The money returned for the hold	VARCHAR(S)	9999			123.93
	Date	The last day of the customer's stay	DATE	mm-dd-yyyy			9/13/2035
	Final Amount	The total amount the customer has to pay	VARCHAR(S)	9999	y		1029.92
Rooms	Room Number	The number of the customer's room	VARCHAR(S)	999	y	PK	485
	Room cleaned?	Is the room cleaned and ready?	CHAR(1)	x	y		y
	Room occupied?	Is the room occupied?	CHAR(1)	x	y		n
	Clock in ID	The staff's clock in ID	VARCHAR(S)	9999			92929
	Room Cost	The cost of room	VARCHAR(S)	99999			92838
	Incidents	Were there any incidents	CHAR(1)	x	y		n
	Avg Room cost	The average cost of the room	VARCHAR(S)	999999			154.39
	Date	The Date of the record	DATE	mm-dd-yyyy	y		10/7/2025

Queries and Operations

Section 1: Entity Generation and Data Entry: Create Database, Create Tables, and Insert Values to the Tables

Overview. Our Hotel Management data model consists of only one hotel database which includes six tables since they are all interconnected.

Table 3: Database Schema

Database	Table Name	Table Explanation	Record
Hotel	Staff	This table contains information about the staff such as clock in ID, what time and date they were clocked in	10
Hotel	Rooms	This table contains information about the rooms that were cleaned or are occupied and who cleaned them.	10
Hotel	Reservation	This table provides information about the customer reservation such as Confirmation Number, date, duration.	10
Hotel	Customer	This table contains customer information such as Credit card number, confirmation number, name.	10
Hotel	Food/Kiosk Sale	This table contains information on what was sold from the kiosk and restaurant. It has columns such as item ID.	10
Hotel	Payment	This table provides information about the final details of the payment.	10

Table 4: Query Steps and Explanations

Steps	Query	Example
Database is created.	CREATE DATABASE DBNAME	CREATE DATABASE hotel
Tables are created.	CREATE TABLE DBNAME.tablename	CREATE TABLE Hotel.Rooms(Room_Number INT NOT NULL, Is_Room_Cleaned CHAR(1) NOT NULL, Is_Room_Occupied CHAR(1) NOT NULL);
Data is inserted into the table.	INSERT INTO DBNAME.TABLENAME VALUES (V1,V2,V3..)	INSERT INTO Hotel.Customer VALUES ('72637378', 7362717273819, 'Mary Stuck',

		'MaryStuck@gmail.com', '7372 7272 8920 0918', 8282, 'Gold', 'Y'),
The values are retrieved.	SELECT from DBname.tablename;	SELECT* from Hotel.Rooms

Entity Generation and Data Entry for Table Staff

Statements Explanation

- First, the database Hotel is created in the database schema. Command "Create DATABASE Hotel " is used to create database.
- Second, the table Staff is created in Hotel database. Command "CREATE TABLE Hotel.Staff Staff " is used to create the table. Command “Use Facility” is used to call the facility database.
- Third, table Staff is filled with relevant data using the command "Insert into”.
- Finally, the result will be displayed by using the command “Select * from Hotel.Staff” to query all the inserted values to the table.

Queries

```
/* --------------------------------------------------Create Hotel Database--------------------------------------------------*/
CREATE DATABASE Hotel

/* -------------------------------------------Create Staff Table-------------------------------------------*/
create TABLE Hotel.Staff (

        Clock_In_ID  CHAR(4) NOT NULL Primary Key,

        Name CHAR(50) NOT NULL,

        Date DATE NOT NULL,

        Time_Clocked_In     TIME NOT NULL,

        Time_Clocked_Out TIME NOT NULL,

        Type CHAR(100) NOT NULL,

        Access_Liquor CHAR(1) NULL,

        Manager CHAR(50) NULL

);

/*----------------------------------------Insert Values to Staff Table-------------------------------------------------*/

INSERT INTO Hotel.Staff
VALUES ('1024','Destiny Archer', '2025-12-04', '08:01:33','17:25:09', 'Housekeeping', 'N',
'Marlon Richie'),
            ('2342','Willow Lover', '2025-12-13', '04:34:24','11:05:30','Breakfast Server', 'N',
'Marlon Richie'),
```

('1834', 'Lorraine Webber', '2025-12-07', '15:30:12','22:37:56', 'Night Cook', 'N', 'Chris Rules'),
('6738', 'Chris Rules', '2025-12-14', '13:57:58','21:23:43', 'Food and Beverage Manager', 'Y', NULL),
('7821', 'Sandra Willis', '2025-12-05', '07:56:12','17:28:52', 'Housekeeping', 'N', 'Marlon Richie'),
('9852', 'Amy Swift', '2025-12-09', '15:02:02','23:02:23', 'Front Desk', 'N', 'Gertrude Plate'),
('7632', 'Gertrude Plate', '2025-12-09', '13:09:43','22:09:10','Front Desk Manager', 'Y', NULL),
('2173', 'Matthew Mccough','2025-12-13', '16:34:55','23:09:21', 'Bartender', 'Y', 'Chris Rules'),
('9821', 'Walter Cunnings', '2025-12-15', '08:09:10','17:19:09', 'Housekeeping', 'N', 'Marlon Richie'),
('0278', 'Milo Manheim', '2025-12-18', '08:05:23', '17:30:34', 'Housekeeping', 'N', 'Marlon Richie');

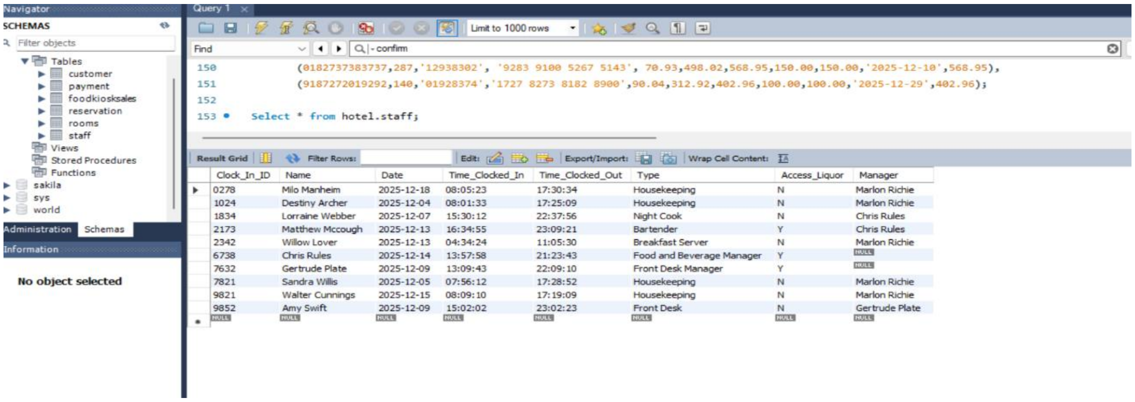
Result

Table Explanation: Table Staff provides information about dates and time clocked in. As well as clock in ID. To query all the information command “Select * from hotel.staff “ is used.

Query

Select * from hotel.staff;

Screenshot



Entity Generation and Data Entry for Table Rooms

Statements Explanation

- The database Hotel is already created.
- The Table ROOMS is created with “CREATE Table Hotel.Rooms”
- Next, the table Rooms is filled with relevant data using the command "Insert into”.
- Finally, the result will be displayed by using the command “Select * from Hotel.Rooms” to query all the inserted values to the table.

Queries

```
/* -------------------------------------------------Create ROOMS Table---------------------------------------------*/
create TABLE Hotel.Rooms (
	Room_Number INT NOT NULL,
	Is_Room_Cleaned CHAR(1) NOT NULL,
	Is_Room_Occupied CHAR(1) NOT NULL,
	Clock_In_ID CHAR(4) NOT NULL,
	Room_Cost DECIMAL(16,2) NOT NULL,
	Are_There_Incidents CHAR(1) NULL,
	Avg_Room_Cost DECIMAL(16,2) NOT NULL,
	Date DATE NOT NULL,
   PRIMARY KEY(Room_Number, Date),
   FOREIGN KEY (Clock_In_ID) REFERENCES Hotel.Staff(Clock_In_ID)
);
/* -------------------------------------------------Insert Values into ROOMS Table----------------------------------------*/
INSERT INTO Hotel.Rooms
VALUES (101, 'Y', 'N', '9821', 290.38,'N',215.92, '2025-12-14'),
            (213, 'N', 'Y', '9821', 140.98, NULL, 125.83, '2025-12-07'),
         (299, 'Y', 'N', '0278', 150.35, 'N', 155.30, '2025-11-09'),
      (207, 'Y', 'N', '7821', 145.98, 'Y', 135.92, '2025-12-10'),
      (150, 'N', 'N','0278', 168.93, 'N', 135.10, '2025-12-02'),
      (132, 'Y', 'Y', '7821', 130.02, NULL, 135.50, '2025-12-02'),
            (209, 'N','N','7821',321.50, 'N',310.92, '2025-12-05'),
      (121, 'N', 'N', '1024', 127.77, 'N', 129.93, '2025-12-10'),
      (287, 'Y', 'N', '1024', 287.09, 'N', 215.92, '2025-12-14'),
      (140, 'N', 'N', '0278', 192.29, 'N', 200.91, '2025-12-13');
```

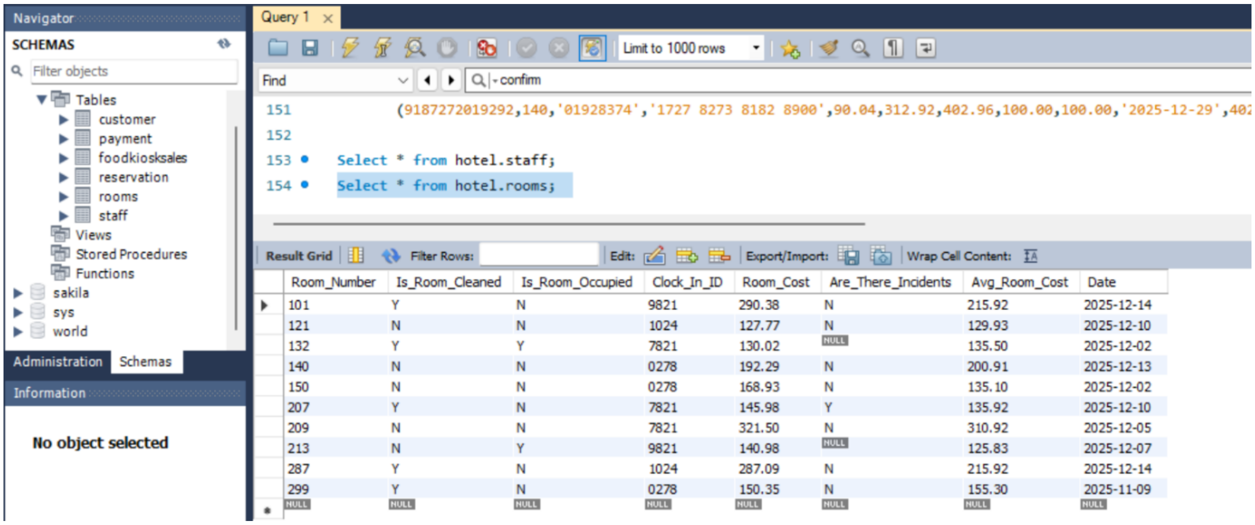
Result

Table Explanation: Table Staff provides information about rooms that are cleaned or occupied, what date the record was made and who was on the shift.

Query

Select * from hotel.rooms;

Screenshot



Room_Number	Is_Room_Cleaned	Is_Room_Occupied	Clock_In_ID	Room_Cost	Are_There_Incidents	Avg_Room_Cost	Date
101	Y	N	9821	290.38	N	215.92	2025-12-14
121	N	N	1024	127.77	N	129.93	2025-12-10
132	Y	Y	7821	130.02	NULL	135.50	2025-12-02
140	N	N	0278	192.29	N	200.91	2025-12-13
150	N	N	0278	168.93	N	135.10	2025-12-02
207	Y	N	7821	145.98	Y	135.92	2025-12-10
209	N	N	7821	321.50	N	310.92	2025-12-05
213	N	Y	9821	140.98	NULL	125.83	2025-12-07
287	Y	N	1024	287.09	N	215.92	2025-12-14
299	Y	N	0278	150.35	N	155.30	2025-11-09
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Entity Generation and Data Entry for Table Reservation

Statements Explanation

- The database Hotel is already created.
- The Table Reservation is created with “CREATE Table Hotel.Reservation”
- Next, the table Reservation is filled with relevant data using the command "Insert into”.
- Finally, the result will be displayed by using the command “Select * from Hotel.Reservation” to query all the inserted values to the table.

Queries

```
/* ----------------------------------------------------Create Reservation Table---------------------------------------------*/
CREATE TABLE Hotel.Reservation (
        Confirmation_Num BIGINT NOT NULL Primary Key,
        Customer_Name CHAR(50) NOT NULL,
        Payment CHAR(10) NOT NULL,
        Type CHAR(10) NULL,
        Room_Number INT NOT NULL ,
        Check_In_Date DATE NOT NULL,
        Check_Out_Date DATE NOT NULL,
        Duration INT NOT NULL,
        Cost DECIMAL(16,2) NOT NULL
);
/* -------------------------------------------Insert Values into Reservation Table--------------------------------------------*/
INSERT INTO Hotel.Reservation
 VALUES (7362717273819, 'Mary Stuck', 'Card', 'Individual', 101, '2025-12-12', '2025-12-15', 3, 892.20),
                (9373717275310, 'Jay Wilson', 'Card', 'Group', 213, '2025-12-06', '2025-12-08', 2, 314.92),
```

(1273678228819, 'Keturah White', 'Cash', 'Individual', 299, '2025-12-09', '2025-12-10', 1,145.20),
(9287465162838, 'Chinedu Opara', 'Card', 'Group',207, '2025-12-06','2025-12-08',2, 278.12),
(9374748838109, 'Kelly Salmon', 'Cash', NULL, 150, '2025-11-30', '2025-12-02',2, 523.09),
(1098273792838, 'Misha Pink', 'Card', 'Individual',132 , '2025-12-03', '2025-12-10', 7, 1493.20),
(0182717273783, 'Breelyn Johnson', 'Cash', 'Group', 209, '2025-12-06','2025-12-08',2, 302.12),
(9019298177374, 'Sarah Lopez', 'Card', 'Individual', 121, '2025-12-23', '2025-12-26', 3, 390.92),
(0182737383737, 'Kathryn Pierce', 'Card', 'Individual', 287, '2025-12-07','2025-12-10', 3, 498.02),
(9187272019292, 'Seonghwa Park', 'Card','Individual',140, '2025-12-27','2025-12-29',2,312.92);

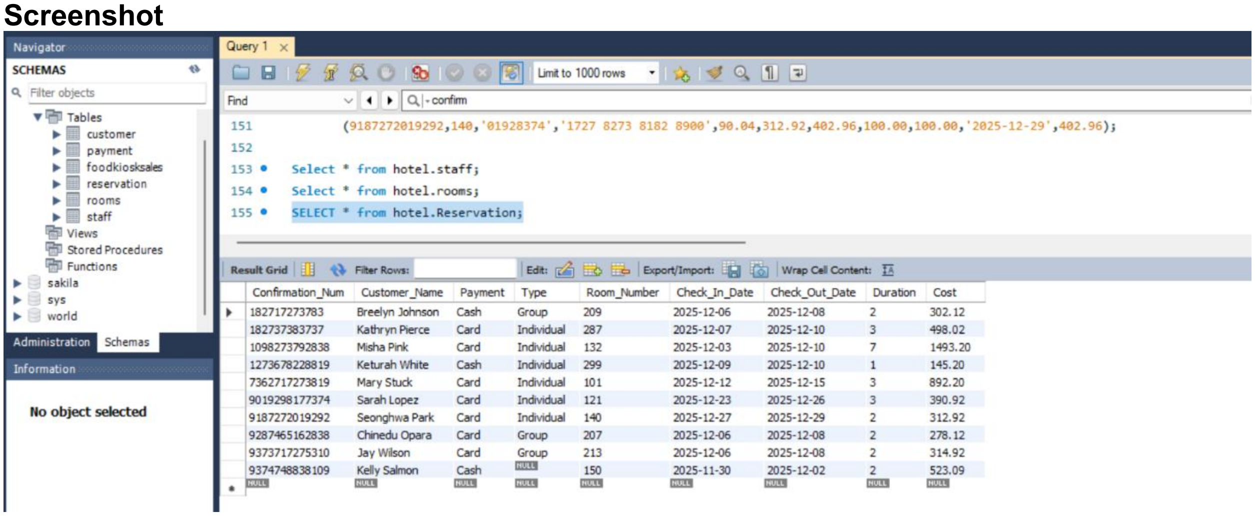
Result

Table Explanation: Table Reservation provides information about the name, confirmation number, room cost, dates for the reservation.

Query

Select * from hotel.Reservation;

Screenshot



Confirmation_Num	Customer_Name	Payment	Type	Room_Number	Check_In_Date	Check_Out_Date	Duration	Cost
182717273783	Breelyn Johnson	Cash	Group	209	2025-12-06	2025-12-08	2	302.12
182737383737	Kathryn Pierce	Card	Individual	287	2025-12-07	2025-12-10	3	498.02
1098273792838	Misha Pink	Card	Individual	132	2025-12-03	2025-12-10	7	1493.20
1273678228819	Keturah White	Cash	Individual	299	2025-12-09	2025-12-10	1	145.20
7362717273819	Mary Stuck	Card	Individual	101	2025-12-12	2025-12-15	3	892.20
9019298177374	Sarah Lopez	Card	Individual	121	2025-12-23	2025-12-26	3	390.92
9187272019292	Seonghwa Park	Card	Individual	140	2025-12-27	2025-12-29	2	312.92
9287465162838	Chinedu Opara	Card	Group	207	2025-12-06	2025-12-08	2	278.12
9373717275310	Jay Wilson	Card	Group	213	2025-12-06	2025-12-08	2	314.92
9374748838109	Kelly Salmon	Cash	NULL	150	2025-11-30	2025-12-02	2	523.09
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Entity Generation and Data Entry for Table Customers

Statements Explanation

- The database Hotel is already created.
- The Table Customers is created with “CREATE Table Hotel.Customer”
- Next, the table Customers is filled with relevant data using the command "Insert into”.

- Finally, the result will be displayed by using the command “Select * from Hotel.Customer” to query all the inserted values to the table.

Queries

```
/* -------------------------------------------------Create Customer Table---------------------------------------------*/
CREATE TABLE Hotel.Customer (
        ID VARCHAR(10) NOT NULL Primary Key,
        Confirmation_Num BIGINT NOT NULL,
        Customer_Name VARCHAR(50) NOT NULL,
        Customer_Email_Address VARCHAR(100) NOT NULL UNIQUE,
        Card_Information VARCHAR(100) NOT NULL UNIQUE,
        Points INT NULL,
        Member_Tier CHAR(50) NULL,
        Team_Member_Or_Family CHAR(1) NULL,
   Foreign Key (Confirmation_Num) references Reservation(Confirmation_Num)

/* ------------------------------------------Insert Values into Customer Table-------------------------------------------*/
INSERT INTO Hotel.Customer
VALUES ('72637378', 7362717273819, 'Mary Stuck', 'MaryStuck@gmail.com', '7372
7272 8920 0918', 8282, 'Gold', 'Y'),
          ('82738129', 9373717275310, 'Jay Wilson', 'JayWilson@yahoo.com', '8278
8171 9091 0987', 1283, 'Silver', 'N'),
     ('01928380', 1273678228819, 'Keturah White', 'KeturahWhite@live.com', '7263
6172 9201 8272', NULL, NULL, NULL),
     ('09172638', 9287465162838, 'Chinedu Opara', 'ChineduOpara@yahoo.com', '8276
7278 9282 8228', 21982, 'Diamond', 'N'),
     ('87738838', 9374748838109, 'Kelly Salmon', 'Kel1208fish@gmail.com', '8283 8281
0098 8793', NULL, NULL, NULL),
     ('91838399', 1098273792838, 'Misha Pink', 'MishaPink1022@yahoo.com', '1029
9292 9281 8238', NULL, NULL, NULL),
     ('82737372', 0182717273783, 'Breelyn Johnson', 'Breelynkitkat@gmail.com', '9283
8282 8282 8282', 1912, 'Silver', 'N'),
          ('72781828', 9019298177374, 'Sarah Lopez', 'SarahLop1993@live.com', '8282
9228 9282 9292', 4120, 'Gold', 'N'),
     ('12938302', 0182737383737, 'Kathryn Pierce', 'KathrynPierce@gmail.com', '9283
9100 5267 5143', NULL,NULL, NULL),
     ('01928374', 9187272019292, 'Seonghwa Park', 'SeonghwaPark@gmail.com',
'1727 8273 8182 8900', 29380, 'Diamond', 'N');
```

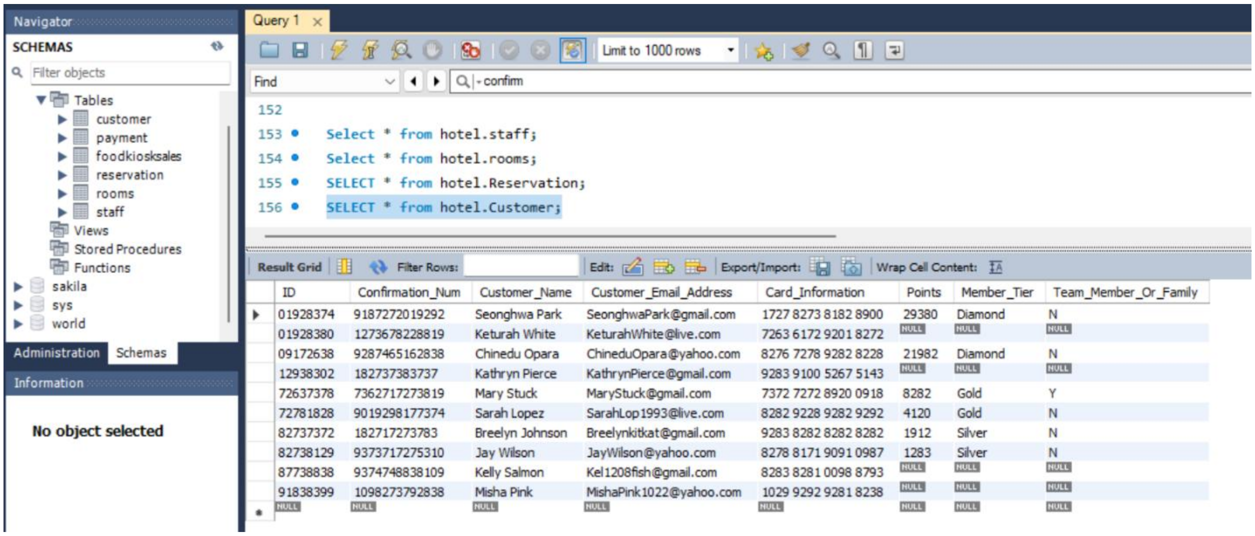
Result

Table Explanation: Customer provides information such as the customer name, what type of member they are, their member tier, their points and card information.

Query

Select * from hotel.Customer;

Screenshot



ID	Confirmation_Num	Customer_Name	Customer_Email_Address	Card_Information	Points	Member_Tier	Team_Member_Or_Family
01928374	9187272019292	Seonghwa Park	SeonghwaPark@gmail.com	1727 8273 8182 8900	29380	Diamond	N
01928380	1273678228819	Keturah White	KeturahWhite@live.com	7263 6172 9201 8272	NULL	NULL	NULL
09172638	9287465162838	Chinedu Opara	ChineduOpara@yahoo.com	8276 7278 9282 8228	21982	Diamond	N
12938302	182737383737	Kathryn Pierce	KathrynPierce@gmail.com	9283 9100 5267 5143	NULL	NULL	NULL
72637378	7362717273819	Mary Stuck	MaryStuck@gmail.com	7372 7272 8920 0918	8282	Gold	Y
72781828	9019298177374	Sarah Lopez	SarahLop1993@live.com	8282 9228 9282 9292	4120	Gold	N
82737372	182717273783	Breelyn Johnson	Breelynkitkat@gmail.com	9283 8282 8282 8282	1912	Silver	N
82738129	9373717275310	Jay Wilson	JayWilson@yahoo.com	8278 8171 9091 0987	1283	Silver	N
87738838	9374748838109	Kelly Salmon	Kel1208fish@gmail.com	8283 8281 0098 8793	NULL	NULL	NULL
91838399	1098273792838	Misha Pink	MishaPink1022@yahoo.com	1029 9292 9281 8238	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Entity Generation and Data Entry for Table FoodKioskSales

Statements Explanation

- The database Hotel is already created.
- The Table Reservation is created with “CREATE Table Hotel.FoodKioskSales”
- Next, the table FoodKioskSales is filled with relevant data using the command "Insert into”.
- Finally, the result will be displayed by using the command “Select * from Hotel.FoodKioskSales” to query all the inserted values to the table.

Queries

```
/* -------------------------------------------------Create FoodKioskSales Table--------------------------------------------*/
create TABLE Hotel.FoodKioskSales (
        Department CHAR(15) NOT NULL,
        Item_Name CHAR(100) NOT NULL,
        Item_ID VARCHAR(10) NOT NULL Primary Key,
        Item_Sale DECIMAL(16,2) NOT NULL,
        Tips DECIMAL(16,2) NULL,
        Date_Of_Purchase DATE NOT NULL,
        Time_of_Purchase TIME NOT NULL,
        Emp_Clock_In_ID   CHAR(4) NULL,
        Room_Number       INT NULL,
   FOREIGN KEY(Emp_Clock_In_ID) REFERENCES Staff(Clock_In_ID),
   FOREIGN KEY(Room_Number) REFERENCES Rooms(Room_Number)
);
```

```
/* ----------------------------------------Insert Values into FoodKioskSales Table----------------------------------------*/
INSERT INTO Hotel.Payment
VALUES (7362717273819, 101,'72637378', '7372 7272 8920 0918', 12.02,
892.20,910.22,150.00,150.00,'2025-12-15',904.44),
            (9373717275310, 213,'82738129','8278 8171 9091 0987',
4.55,314.92,319.47,100.00,100.00,'2025-12-08',319.47),
      (1273678228819, 299,'01928380', '7263 6172 9201
8272',45.93,145.20,191.13,50.00,50.00,'2025-12-10',191.13),
      (9287465162838,207,'09172638', '8276 7278 9282 8228', 33.02,
278.12,311.14,100.00,100.00,'2025-12-08',311.14),
      (9374748838109, 150,'87738838', '8283 8281 0098 8793',NULL,
523.09,523.09,100.00,100.00,'2025-12-02',523.09),
      (1098273792838, 132,'91838399','1029 9292 9281
8238',125.02,1493.20,1618.22,350.00,350.00,'2025-12-10',1618.22),
      (0182717273783, 209,'82737372', '9283 8282 8282
8282',NULL,302.12,302.12,100.00,100.00,'2025-12-08',301.12),
      (9019298177374,121,'72781828', '8282 9228 9282 9292',
11.17,390.92,402.09,150.00,150.00,'2025-12-26',402.09),
      (0182737383737,287,'12938302', '9283 9100 5267 5143',
70.93,498.02,568.95,150.00,150.00,'2025-12-10',568.95),
            (9187272019292,140,'01928374','1727 8273 8182
8900',90.04,312.92,402.96,100.00,100.00,'2025-12-29',402.96);
```

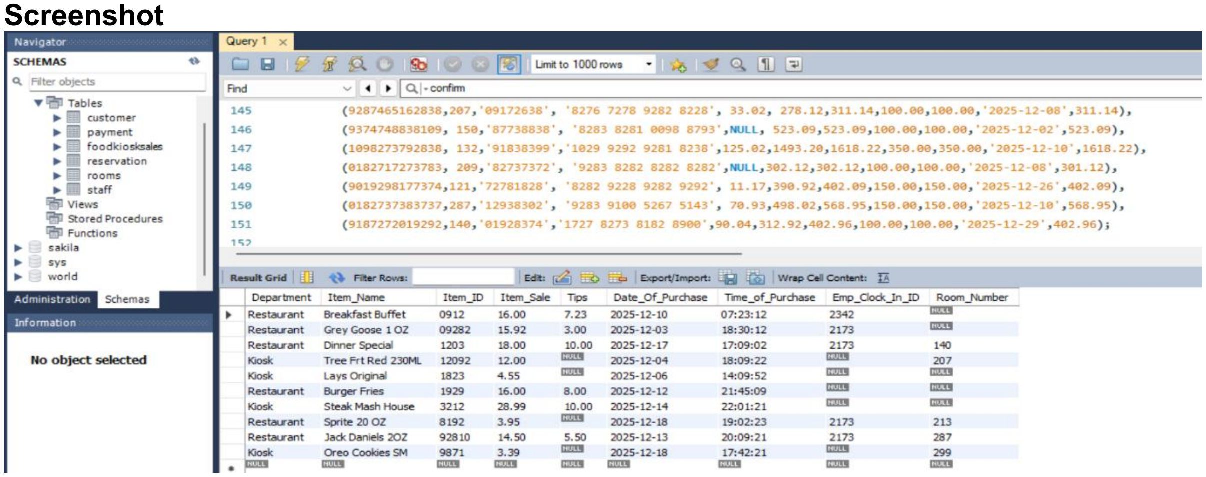
Result

Table Explanation: Food Kiosk Sales provide information about the food, sales, tips made on the food and drinks as well as the employee id.

Query

Select * from hotel.FoodKioskSales;

Screenshot



Department	Item_Name	Item_ID	Item_Sale	Tips	Date_Of_Purchase	Time_of_Purchase	Emp_Clock_In_ID	Room_Number
Restaurant	Breakfast Buffet	0912	16.00	7.23	2025-12-10	07:23:12	2342	NULL
Restaurant	Grey Goose 1 OZ	09282	15.92	3.00	2025-12-03	18:30:12	2173	NULL
Restaurant	Dinner Special	1203	18.00	10.00	2025-12-17	17:09:02	2173	140
Kiosk	Tree Frt Red 230ML	12092	12.00	NULL	2025-12-04	18:09:22	NULL	207
Kiosk	Lays Original	1823	4.55	NULL	2025-12-06	14:09:52	NULL	NULL
Restaurant	Burger Fries	1929	16.00	8.00	2025-12-12	21:45:09	NULL	NULL
Kiosk	Steak Mash House	3212	28.99	10.00	2025-12-14	22:01:21	NULL	NULL
Restaurant	Sprite 20 OZ	8192	3.95	NULL	2025-12-18	19:02:23	2173	213
Restaurant	Jack Daniels 2OZ	92810	14.50	5.50	2025-12-13	20:09:21	2173	287
Kiosk	Oreo Cookies SM	9871	3.39	NULL	2025-12-18	17:42:21	NULL	299
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Entity Generation and Data Entry for Table FoodKioskSales

Statements Explanation

- The database Hotel is already created.
- The Table Reservation is created with “CREATE Table Hotel.Payment”
- Next, the table Payment is filled with relevant data using the command "Insert into”.
- Finally, the result will be displayed by using the command “Select * from Hotel.Payment” to query all the inserted values to the table.

Queries

```
/* ------------------------------------------------Create Payment Table--------------------------------------------*/
create TABLE Hotel.Payment (
        Confirmation_Num BIGINT NOT NULL,
        Room_Number       INT NOT NULL,
   ID VARCHAR(10) NOT NULL,
        Card_Information VARCHAR(100) NOT NULL,
        Food_Kiosk_Sale DECIMAL(16,2) NULL,
        Room_Cost DECIMAL(16,2) NOT NULL,
        Payment_Due DECIMAL(16,2) NOT NULL,
        Incidental_Hold       DECIMAL(16,2) NOT NULL,
        Money_Returned_To_Credit_Card         DECIMAL(16,2) NULL,
        Date DATE NOT NULL,
        Final_Amount DECIMAL(16,2) NOT NULL,
   FOREIGN KEY (Confirmation_Num) REFERENCES
Hotel.Reservation(Confirmation_Num),
   FOREIGN KEY (Room_Number) References Rooms(Room_Number),
   PRIMARY KEY (Confirmation_Num,ID)
);

/* --------------------------------------------Insert Values Into Payment Table--------------------------------------------*/
INSERT INTO Hotel.Payment
VALUES (7362717273819, 101,'72637378', '7372 7272 8920 0918', 12.02,
892.20,910.22,150.00,150.00,'2025-12-15',904.44),
            (9373717275310, 213,'82738129','8278 8171 9091 0987',
4.55,314.92,319.47,100.00,100.00,'2025-12-08',319.47),
      (1273678228819, 299,'01928380', '7263 6172 9201
8272',45.93,145.20,191.13,50.00,50.00,'2025-12-10',191.13),
      (9287465162838,207,'09172638', '8276 7278 9282 8228', 33.02,
278.12,311.14,100.00,100.00,'2025-12-08',311.14),
      (9374748838109, 150,'87738838', '8283 8281 0098 8793',NULL,
523.09,523.09,100.00,100.00,'2025-12-02',523.09),
      (1098273792838, 132,'91838399','1029 9292 9281
8238',125.02,1493.20,1618.22,350.00,350.00,'2025-12-10',1618.22),
      (0182717273783, 209,'82737372', '9283 8282 8282
8282',NULL,302.12,302.12,100.00,100.00,'2025-12-08',301.12),
      (9019298177374,121,'72781828', '8282 9228 9282 9292',
11.17,390.92,402.09,150.00,150.00,'2025-12-26',402.09),
```

(0182737383737,287,'12938302', '9283 9100 5267 5143', 70.93,498.02,568.95,150.00,150.00,'2025-12-10',568.95),
(9187272019292,140,'01928374','1727 8273 8182 8900',90.04,312.92,402.96,100.00,100.00,'2025-12-29',402.96);

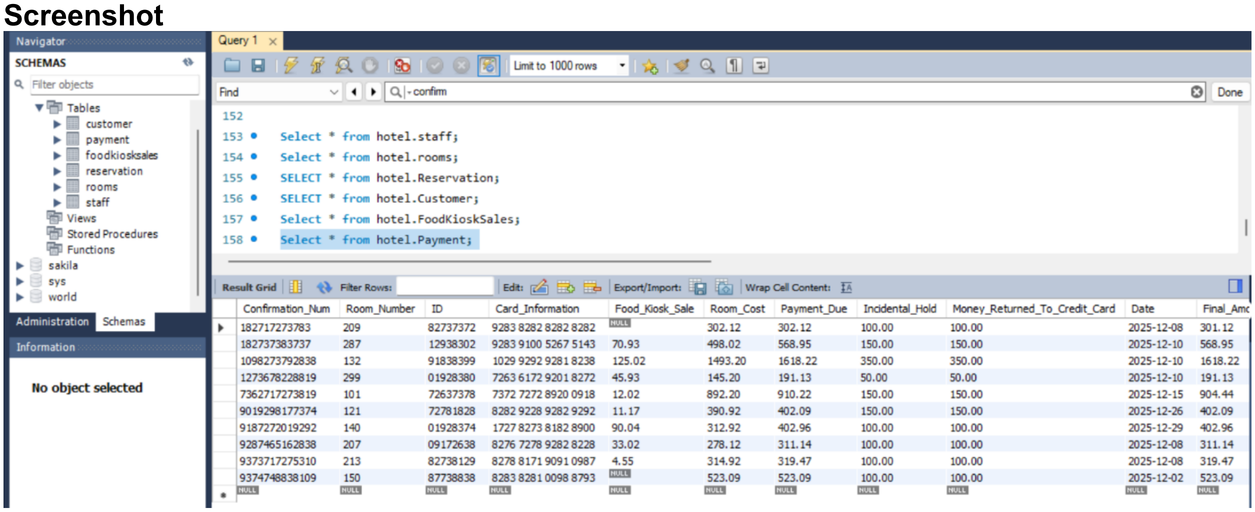
Result

Table Explanation: Payment Table provides information about the final payments due from the customer and the money that needs to be returned on their credit card for incidentals.

Query

Select * from hotel.Payment;

Screenshot



Data Analysis 1

Statement: How many customers stayed more than 2 days during their trip?

SQL Command Explanation:

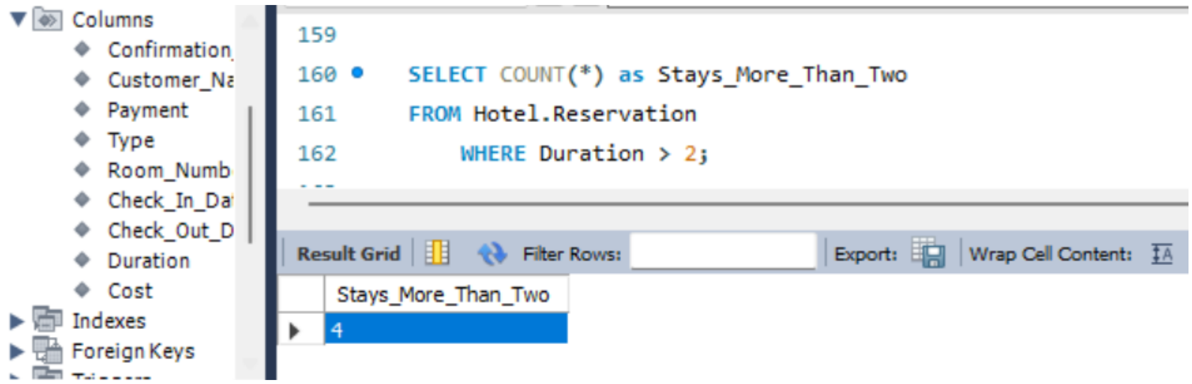
- Command FROM to retrieve the data from the Reservation table in the Hotel Database.
- Command WHERE to filter the data to meet certain criteria, duration >2.
- Command SELECT to show the results from certain columns with count to count the number of rows.

SQL Query:

```
/*---------------------------------------------------- Data Analysis 1:----------------------------------------*/

SELECT COUNT(*) as Stays_More_Than_Two
FROM Hotel.Reservation
        WHERE Duration > 2;
```

Result and Screenshot:



Data Analysis 2

Statement: Calculate the total amount spent by each member tier.

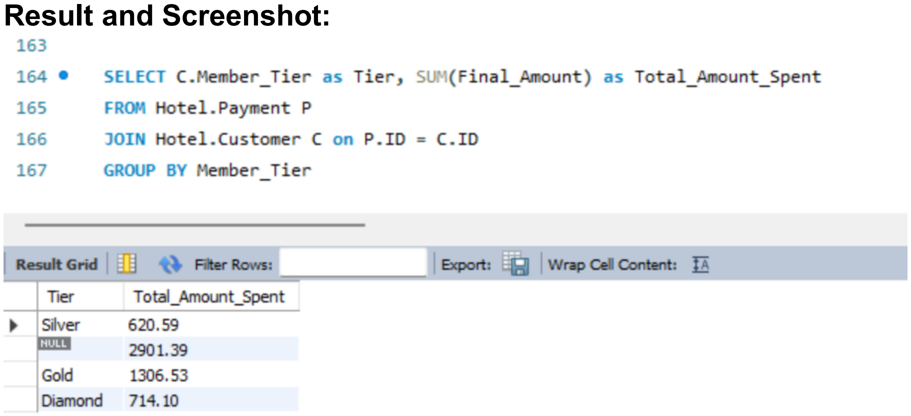
SQL Command Explanation:

- Command FROM to retrieve the data from the Payment P table in the Hotel Database.
- Commandt JOIN Hotel.Customer C to join the tables.
- Command SELECT to show the results from certain columns with SUM of final Amount.
- Command GROUP BY to group the data by Member_Tier

SQL Query:

```
/*------------------------------------------------------- Data Analysis 2:-------------------------------------------*/
SELECT C.Member_Tier as Tier, SUM(Final_Amount) as Total_Amount_Spent
FROM Hotel.Payment P
JOIN Hotel.Customer C
        on P.ID = C.ID
GROUP BY Member_Tier
```

Result and Screenshot:



Data Analysis 3

Statement: At what time is the most food/kiosk sales made?

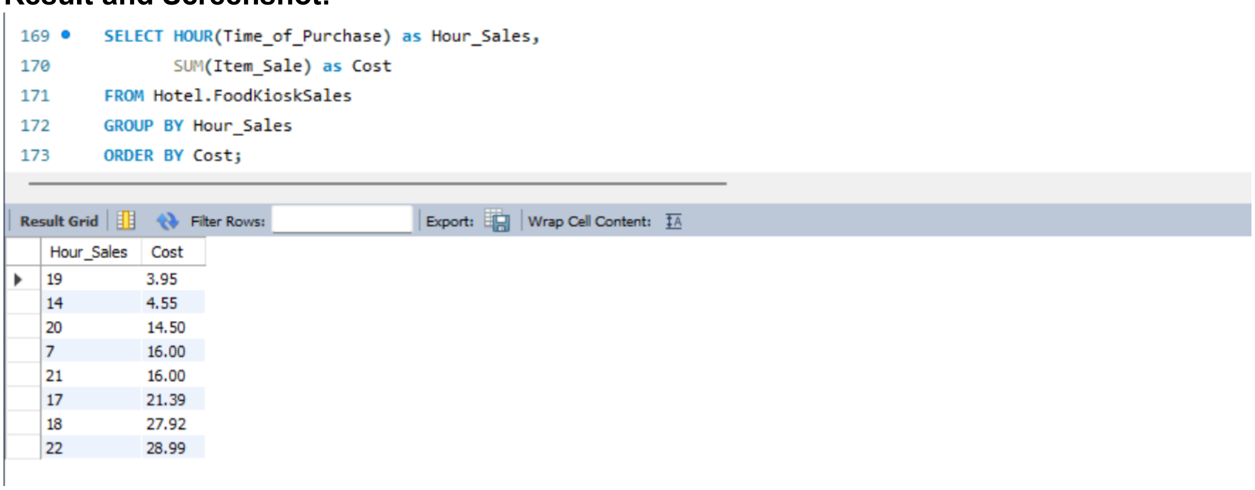
SQL Command Explanation:

- Command FROM to retrieve the data from the FoodKioskSales table in the Hotel Database.
- Command SELECT to show the results from certain columns with SUM of Item_Cost as Cost and Hour(Time_of_Purchase) as Hour_Sales
- Command GROUP BY to group the data by Time of Sales
- Command Order by Cost

SQL Query:

```
/*---------------------------------------------------- Data Analysis 3:-----------------------------------------*/
SELECT HOUR(Time_of_Purchase) as Hour_Sales,
            SUM(Item_Sale) as Cost
FROM Hotel.FoodKioskSales
GROUP BY Hour_Sales
ORDER BY Cost;
```

Result and Screenshot:



```
SELECT HOUR(Time_of_Purchase) as Hour_Sales,
       SUM(Item_Sale) as Cost
FROM Hotel.FoodKioskSales
GROUP BY Hour_Sales
ORDER BY Cost;
```


Hour_Sales	Cost
19	3.95
14	4.55
20	14.50
7	16.00
21	16.00
17	21.39
18	27.92
22	28.99

Data Analysis 4

Statement: Which customers had points higher or equal to1500 and had final payment over 300?

SQL Command Explanation:

- Command FROM to retrieve the data from the Reservation table in the Hotel Database.
- Command JOIN Reservation R to Customer C to Payment P
- Command SELECT to show the results from certain columns with Customer_Name, Sum of Final Amount, Points
- Command GROUP BY to group the data by Customer_Name and Points
- Command Order by Total_Amount_Spent

SQL Query:

/*-- Data Analysis 4:--*/
SELECT C.Customer_Name as Names,C.Points as Points, Sum(P.Food_Kiosk_Sale +
P.Room_Cost) as Total_Amount_Spent
FROM Hotel.Reservation R
JOIN Hotel.Customer C
on R.Confirmation_Num = C.Confirmation_Num
JOIN Hotel.Payment P
on C.ID = P.ID
GROUP BY Names,Points
ORDER BY Total_Amount_Spent

Result and Screenshot:

```
SELECT C.Customer_Name as Names,C.Points as Points, Sum(P.Food_Kiosk_Sale + P.Room_Cost) as Total_Amount_Spent
FROM Hotel.Reservation R
JOIN Hotel.Customer C
on R.Confirmation_Num = C.Confirmation_Num
JOIN Hotel.Payment P
on C.ID = P.ID
GROUP BY Names,Points
ORDER BY Total_Amount_Spent
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Names	Points	Total_Amount_Spent
Breelyn Johnson	1912	NULL
Kelly Salmon	NULL	NULL
Keturah White	NULL	191.13
Chinedu Opara	21982	311.14
Jay Wilson	1283	319.47
Sarah Lopez	4120	402.09
Seonghwa Park	29380	402.96
Kathryn Pierce	NULL	568.95
Mary Stuck	8282	904.22
Misha Pink	NULL	1618.22

Appendix 1

/* ---Create Hotel Database--*/

```
CREATE DATABASE Hotel

/* --------------------------------------------Create Staff Table--------------------------------------------*/
create TABLE Hotel.Staff (
          Clock_In_ID  CHAR(4) NOT NULL Primary Key,
          Name CHAR(50) NOT NULL,
          Date DATE NOT NULL,
          Time_Clocked_In     TIME NOT NULL,
          Time_Clocked_Out TIME NOT NULL,
          Type CHAR(100) NOT NULL,
          Access_Liquor CHAR(1) NULL,
          Manager CHAR(50) NULL

/* --------------------------------------------------Create ROOMS Table--------------------------------------------*/
create TABLE Hotel.Rooms (
          Room_Number INT NOT NULL,
          Is_Room_Cleaned CHAR(1) NOT NULL,
          Is_Room_Occupied CHAR(1) NOT NULL,
          Clock_In_ID CHAR(4) NOT NULL,
          Room_Cost DECIMAL(16,2) NOT NULL,
          Are_There_Incidents CHAR(1) NULL,
          Avg_Room_Cost DECIMAL(16,2) NOT NULL,
          Date DATE NOT NULL,
   PRIMARY KEY(Room_Number, Date),
   FOREIGN KEY (Clock_In_ID) REFERENCES Hotel.Staff(Clock_In_ID)
);

/* --------------------------------------------------Create Reservation Table--------------------------------------------*/
CREATE TABLE Hotel.Reservation (
          Confirmation_Num BIGINT NOT NULL Primary Key,
          Customer_Name CHAR(50) NOT NULL,
          Payment CHAR(10) NOT NULL,
          Type CHAR(10) NULL,
          Room_Number INT NOT NULL ,
          Check_In_Date DATE NOT NULL,
          Check_Out_Date DATE NOT NULL,
          Duration INT NOT NULL,
          Cost DECIMAL(16,2) NOT NULL

/* --------------------------------------------------Create Customer Table--------------------------------------------*/
CREATE TABLE Hotel.Customer (
          ID VARCHAR(10) NOT NULL Primary Key,
          Confirmation_Num BIGINT NOT NULL,
          Customer_Name VARCHAR(50) NOT NULL,
          Customer_Email_Address VARCHAR(100) NOT NULL UNIQUE,
          Card_Information VARCHAR(100) NOT NULL UNIQUE,
          Points INT NULL,
          Member_Tier CHAR(50) NULL,
          Team_Member_Or_Family CHAR(1) NULL,
```

```
   Foreign Key (Confirmation_Num) references Reservation(Confirmation_Num)
);

/* ------------------------------------------------Create FoodKioskSales Table-------------------------------------------*/
create TABLE Hotel.FoodKioskSales (
        Department CHAR(15) NOT NULL,
        Item_Name CHAR(100) NOT NULL,
        Item_ID VARCHAR(10) NOT NULL Primary Key,
        Item_Sale DECIMAL(16,2) NOT NULL,
        Tips DECIMAL(16,2) NULL,
        Date_Of_Purchase DATE NOT NULL,
        Time_of_Purchase TIME NOT NULL,
        Emp_Clock_In_ID   CHAR(4) NULL,
        Room_Number       INT NULL,
   FOREIGN KEY(Emp_Clock_In_ID) REFERENCES Staff(Clock_In_ID),
   FOREIGN KEY(Room_Number) REFERENCES Rooms(Room_Number)
);

/* ------------------------------------------------Create Payment Table-------------------------------------------*/
create TABLE Hotel.Payment (
        Confirmation_Num BIGINT NOT NULL,
        Room_Number       INT NOT NULL,
   ID VARCHAR(10) NOT NULL,
        Card_Information VARCHAR(100) NOT NULL,
        Food_Kiosk_Sale DECIMAL(16,2) NULL,
        Room_Cost DECIMAL(16,2) NOT NULL,
        Payment_Due DECIMAL(16,2) NOT NULL,
        Incidental_Hold       DECIMAL(16,2) NOT NULL,
        Money_Returned_To_Credit_Card         DECIMAL(16,2) NULL,
        Date DATE NOT NULL,
        Final_Amount DECIMAL(16,2) NOT NULL,
   FOREIGN KEY (Confirmation_Num) REFERENCES
Hotel.Reservation(Confirmation_Num),
   FOREIGN KEY (Room_Number) References Rooms(Room_Number),
   PRIMARY KEY (Confirmation_Num,ID)
);

/*------------------------------------------Insert Values to Staff Table-------------------------------------------------*/

INSERT INTO Hotel.Staff
VALUES ('1024','Destiny Archer', '2025-12-04', '08:01:33','17:25:09', 'Housekeeping', 'N',
'Marlon Richie'),
          ('2342','Willow Lover', '2025-12-13', '04:34:24','11:05:30','Breakfast Server', 'N',
'Marlon Richie'),
     ('1834', 'Lorraine Webber', '2025-12-07', '15:30:12','22:37:56', 'Night Cook', 'N',
'Chris Rules'),
```

```
    ('6738', 'Chris Rules', '2025-12-14', '13:57:58','21:23:43', 'Food and Beverage
Manager', 'Y', NULL),
    ('7821', 'Sandra Willis', '2025-12-05', '07:56:12','17:28:52', 'Housekeeping', 'N',
'Marlon Richie'),
    ('9852', 'Amy Swift', '2025-12-09', '15:02:02','23:02:23', 'Front Desk', 'N', 'Gertrude
Plate'),
    ('7632', 'Gertrude Plate', '2025-12-09', '13:09:43','22:09:10','Front Desk Manager',
'Y', NULL),
    ('2173', 'Matthew Mccough','2025-12-13', '16:34:55','23:09:21', 'Bartender', 'Y',
'Chris Rules'),
    ('9821', 'Walter Cunnings', '2025-12-15', '08:09:10','17:19:09', 'Housekeeping', 'N',
'Marlon Richie'),
    ('0278', 'Milo Manheim', '2025-12-18', '08:05:23', '17:30:34', 'Housekeeping', 'N',
'Marlon Richie');
/*----------------------------------------Insert Values to Rooms Table-------------------------------------------------*/

INSERT INTO Hotel.Rooms
VALUES (101, 'Y', 'N', '9821', 290.38,'N',215.92, '2025-12-14'),
        (213, 'N', 'Y', '9821', 140.98, NULL, 125.83, '2025-12-07'),
      (299, 'Y', 'N', '0278', 150.35, 'N', 155.30, '2025-11-09'),
    (207, 'Y', 'N', '7821', 145.98, 'Y', 135.92, '2025-12-10'),
    (150, 'N', 'N','0278', 168.93, 'N', 135.10, '2025-12-02'),
    (132, 'Y', 'Y', '7821', 130.02, NULL, 135.50, '2025-12-02'),
        (209, 'N','N','7821',321.50, 'N',310.92, '2025-12-05'),
    (121, 'N', 'N', '1024', 127.77, 'N', 129.93, '2025-12-10'),
    (287, 'Y', 'N', '1024', 287.09, 'N', 215.92, '2025-12-14'),
    (140, 'N', 'N', '0278', 192.29, 'N', 200.91, '2025-12-13');

/* -----------------------------------------Insert Values into Reservation Table-------------------------------------------*/
INSERT INTO Hotel.Reservation
 VALUES (7362717273819, 'Mary Stuck', 'Card', 'Individual', 101, '2025-12-12', '2025-
12-15', 3, 892.20),
            (9373717275310, 'Jay Wilson', 'Card', 'Group', 213, '2025-12-06', '2025-
12-08', 2, 314.92),
    (1273678228819, 'Keturah White', 'Cash', 'Individual', 299, '2025-12-09', '2025-12-
10', 1,145.20),
    (9287465162838, 'Chinedu Opara', 'Card', 'Group',207, '2025-12-06','2025-12-
08',2, 278.12),
    (9374748838109, 'Kelly Salmon', 'Cash', NULL, 150, '2025-11-30', '2025-12-02',2,
523.09),
    (1098273792838, 'Misha Pink', 'Card', 'Individual',132 , '2025-12-03', '2025-12-10',
7, 1493.20),
    (0182717273783, 'Breelyn Johnson', 'Cash', 'Group', 209, '2025-12-06','2025-12-
08',2, 302.12),
    (9019298177374, 'Sarah Lopez', 'Card', 'Individual', 121, '2025-12-23', '2025-12-
26', 3, 390.92),
```

```
      (0182737383737, 'Kathryn Pierce', 'Card', 'Individual', 287, '2025-12-07','2025-12-
10', 3, 498.02),
      (9187272019292, 'Seonghwa Park', 'Card','Individual',140, '2025-12-27','2025-12-
29',2,312.92);

/* ----------------------------------------Insert Values into FoodKioskSales Table----------------------------------------*/
INSERT INTO Hotel.Payment
VALUES (7362717273819, 101,'72637378', '7372 7272 8920 0918', 12.02,
892.20,910.22,150.00,150.00,'2025-12-15',904.44),
            (9373717275310, 213,'82738129','8278 8171 9091 0987',
4.55,314.92,319.47,100.00,100.00,'2025-12-08',319.47),
      (1273678228819, 299,'01928380', '7263 6172 9201
8272',45.93,145.20,191.13,50.00,50.00,'2025-12-10',191.13),
      (9287465162838,207,'09172638', '8276 7278 9282 8228', 33.02,
278.12,311.14,100.00,100.00,'2025-12-08',311.14),
      (9374748838109, 150,'87738838', '8283 8281 0098 8793',NULL,
523.09,523.09,100.00,100.00,'2025-12-02',523.09),
      (1098273792838, 132,'91838399','1029 9292 9281
8238',125.02,1493.20,1618.22,350.00,350.00,'2025-12-10',1618.22),
      (0182717273783, 209,'82737372', '9283 8282 8282
8282',NULL,302.12,302.12,100.00,100.00,'2025-12-08',301.12),
      (9019298177374,121,'72781828', '8282 9228 9282 9292',
11.17,390.92,402.09,150.00,150.00,'2025-12-26',402.09),
      (0182737383737,287,'12938302', '9283 9100 5267 5143',
70.93,498.02,568.95,150.00,150.00,'2025-12-10',568.95),
            (9187272019292,140,'01928374','1727 8273 8182
8900',90.04,312.92,402.96,100.00,100.00,'2025-12-29',402.96);

/* ---------------------------------------------Insert Values Into Payment Table---------------------------------------------*/
INSERT INTO Hotel.Payment
VALUES (7362717273819, 101,'72637378', '7372 7272 8920 0918', 12.02,
892.20,910.22,150.00,150.00,'2025-12-15',904.44),
            (9373717275310, 213,'82738129','8278 8171 9091 0987',
4.55,314.92,319.47,100.00,100.00,'2025-12-08',319.47),
      (1273678228819, 299,'01928380', '7263 6172 9201
8272',45.93,145.20,191.13,50.00,50.00,'2025-12-10',191.13),
      (9287465162838,207,'09172638', '8276 7278 9282 8228', 33.02,
278.12,311.14,100.00,100.00,'2025-12-08',311.14),
      (9374748838109, 150,'87738838', '8283 8281 0098 8793',NULL,
523.09,523.09,100.00,100.00,'2025-12-02',523.09),
      (1098273792838, 132,'91838399','1029 9292 9281
8238',125.02,1493.20,1618.22,350.00,350.00,'2025-12-10',1618.22),
      (0182717273783, 209,'82737372', '9283 8282 8282
8282',NULL,302.12,302.12,100.00,100.00,'2025-12-08',301.12),
```